TECHNIQUE MASTERING EXERCISES

N INTEGRATION BY SUBSTITUTION

FOR THREE OF THE INTEGRALS YOU HAVE TO DO LONG DIVISION WITH POLYNOMIALS. CANNOT DO IT? CANNOT REMEMBER HOW TO DO IT? YOU WILL DO IT AGAIN IN THE SECOND SEMESTER. I WILL NOT ASK SUCH A PROBLEM FOR THE EXAM.

1. $\int \frac{x}{\sqrt{x+9}}\,dx$

Let $\sqrt{x+9} = u$ then $x+9 = u^2 \Rightarrow x = u^2 - 9 \Rightarrow \frac{dx}{du} = 2u$

$$\int \frac{x}{\sqrt{x+9}}\,dx = \int \frac{u^2-9}{u}(2u)\,du = 2\int (u^2-9)\,du = \tfrac{2}{3}u^3 - 18u + c = \tfrac{2}{3}(x+9)^{\frac{3}{2}} - 18\sqrt{x+9}$$

2. $\int \frac{\sqrt{x-1}}{x}\,dx$:

Let $\sqrt{x-1} = u$ then $x-1 = u^2 \Rightarrow x = u^2+1 \Rightarrow \frac{dx}{du} = 2u$

$$\int \frac{\sqrt{x-1}}{x}\,dx = \int \frac{u}{u^2+1}(2u)\,du = 2\int \frac{u^2}{u^2+1}\,du$$

$$= 2\int \frac{(u^2+1)-1}{u^2+1}\,du = 2\int 1 - \frac{1}{u^2+1}\,du = 2(u - \arctan u) + c$$

$$= 2\sqrt{(x-1)} - 2\arctan\sqrt{(x-1)} + c$$

3. $\int \frac{x}{\sqrt{x}+1}\,dx$

Let $\sqrt{x} = u$ then $x = u^2 \Rightarrow \frac{dx}{du} = 2u$

$$\int \frac{x}{\sqrt{x}+1}\,dx = \int \frac{u^2}{u+1}(2u)\,du = 2\int \frac{u^3}{u+1}\,du$$

$$= 2\int u^2 - u + 1 - \frac{1}{u+1})\,du \text{ (using division)}$$

$$= 2[\tfrac{1}{3}u^3 - \tfrac{1}{2}u^2 + u - \ln|u+1| + c = \tfrac{2}{3}x^{\frac{3}{2}} - x + 2\sqrt{x} - 2\ln|\sqrt{x}+1| + c$$

4. $\int \frac{\sqrt[3]{x}}{\sqrt[3]{x}+1}\,dx$

Let $\sqrt[3]{x} = u$ then $x = u^3 \Rightarrow \frac{dx}{du} = 3u^2$

$$\int \frac{\sqrt[3]{x}}{\sqrt[3]{x}+1}\,dx = \int \frac{u}{u+1}(3u^2)dx = 3\int \frac{u^3}{u+1}\,dx$$

$$= 3\int u^2 - u + 1 - \frac{1}{u+1})\,du \text{ (using division)}$$

$$= 3[\tfrac{1}{3}u^3 - \tfrac{1}{2}u^2 + u - \ln|u+1| + c = x - \tfrac{3}{2}x^{\frac{2}{3}} + 3\sqrt[3]{x} - 3\ln|\sqrt[3]{x}+1| + c$$